Logic, Sets and Numbers

Logic

Def. 1. *A* ***mathematical Statement*** *is a well-formulated assertion that is strictly either true or false.*

Def. 2. *A* ***Statement*** *that depends on one or multiple variables is called a* ***predicate*** *and denoted by $P(x), Q(x), \dots$.*

Def. 3. ***Quantifiers*** *are short hand notation for making statements about predicates.*

$$\forall x \in \mathbb{R} : P(x) \iff \text{for all} x \in \mathbb{R} \text{ holds } P(x)$$

$$\exists x \in \mathbb{R} : P(x) \iff \text{there exists} x \in \mathbb{R} \text{ s.t. } P(x)$$

$$\exists! x \in \mathbb{R} : P(x) \iff \text{there exists exactly one} x \ldots$$

$$\nexists x \in \mathbb{R} : P(x) \iff \text{there exists no} x \in \mathbb{R} \text{ s.t. } \ldots$$

Imp. 4. *The order of quantifiers matters. $\forall x \exists y$ is not equivalent to $\exists y \forall x$.*

Def. 5. *Basic* ***logical connectives*** *are defined as:*

$$A \land B := A \text{ and } B \text{ are true}$$

$$A \lor B := A \text{ or } B \text{ is true}$$

$$A \implies B := A \text{ implies } B \text{ is true}$$

$$A \iff B := A \text{ is equivalent to } B \text{ is true}$$

Lem. 6.

$$\neg A \lor B \iff A \implies B$$

Thm. 7. *The following equivalences hold:*

$$\neg(\forall x P(x)) \iff \exists x \neg P(x)$$

$$\neg(\exists x P(x)) \iff \forall x \neg P(x)$$

aswell as

$$\forall x (P(x) \land Q(x)) \iff (\forall x P(x)) \land (\forall x Q(x))$$

$$\exists x (P(x) \lor Q(x)) \iff (\exists x P(x)) \lor (\exists x Q(x))$$

Sets

Def. 8. *A* ***set*** *is a collection of elements, that are listed or characterized by a certain property. The empty set $:= \emptyset$ and is the only set that contains no elements.*

Def. 9. *Let A and B be sets. A is a* ***subset*** *of B denoted by $A \subset B$ iff.*

$$\forall x (x \in A \implies x \in B)$$

A is a ***proper subset*** *of B denoted by $A \subsetneq B$ iff.*

$$A \subset B \land A \neq B$$

Def. 10. *Let A and B be sets. The* ***union*** *of A and $B := A \cup B := \{x \mid x \in A \lor x \in B\}$.*

Def. 11. *The* ***intersection*** *of A and $B := A \cap B := \{x \mid x \in A \land x \in B\}$.*

Def. 12. *The* ***difference*** *of A and $B := A \setminus B := \{x \mid x \in A \land x \notin B\}$.*

Def. 13. *The* ***Cartesian product*** *of A and B $:= A \times B := \{(a, b) \mid a \in A \land b \in B\}$.*

Rem. 14. *Note that if $B \subset A$ then the diffrence $A \setminus B$ is called the complement of B in A and $:= B^c$.*

Numbers

Rem. 15. *Some important subsets of the real numbers are*

$$\mathbb{R}_0^+ / \mathbb{R}^+$$

$$\mathbb{R}_0^- / \mathbb{R}^-$$

which define the intervalls

$$[0, \infty)/(0, \infty)$$

$$(-\infty, 0]/(-\infty, 0)$$

Def. 16. *Continuous subsets are called* ***intervalls.***

Def. 17. *An intervall is called* ***open*** *if it does not contain its endpoints*

$$I = (a, b) := \{x \mid a < x < b\}$$

closed *if it contains its endpoints*

$$I = [a, b] := \{x \mid a \leq x \leq b\}$$

half-open *if it contains one endpoint and not the other*

$$I = [a, b) := \{x \mid a \leq x < b\}$$

Rem. 18. *An arbitrary subset of $\mathbb{R}$ is called open if it is a union of open intervalls and closed if its complement is open.*

Def. 19. *An* ***upper (lower) bound*** *of a subset $X \subset \mathbb{R}$ is an element $y \in \mathbb{R}$ such that:*

$$\forall x \in X : x \leq (\geq) y$$

The ***maximum (minimum)*** *of a subset $X \subset \mathbb{R}$ is the element $x_0 \in X$ such that:*

$$\forall x \in X : x \leq (\geq) x_0$$

Often denoted as ***max(X)*** *and* ***min(X)***.

Def. 20. *An intervall $I = (a, b)$ is* ***bounded*** *if there exist upper and lower bounds hence*

$$a > -\infty \quad \text{and} \quad b < \infty$$

Rem. 21. *A bounded and closed intervall is often refered to as* ***compact.***

Def. 22. *The* ***Supremum*** *$sup(X)$ of a subset $X \subset \mathbb{R}$ is the least upper bound of X. and can be characterized by:*

$$\forall x \in X : x \leq sup(X) \land$$

$$\forall \epsilon > 0 : \exists x \in X : x > sup(X) - \epsilon$$

Def. 23. *The* ***Infimum*** *$inf(X)$ of a subset $X \subset \mathbb{R}$ is the greatest lower bound of X. and can be characterized by:*

$$\forall x \in X : x \geq inf(X) \land$$

$$\forall \epsilon > 0 : \exists x \in X : x < inf(X) + \epsilon$$

Thm. 24. *The* ***maximum and minimum are unique*** *aslong as they exist.*

Def. 25. ***Order Axioms*** *reflexivity:*

$$\forall x \in X : x \leq x$$

antisymmetry:

$$\forall x, y \in X : x \leq y \land y \leq x \implies x = y$$

transitivity:

$$\forall x, y, z \in X : x \leq y \land y \leq z \implies x \leq z$$

totality:

$$\forall x, y \in X : x \leq y \lor y \leq x$$

Def. 26. *A set X is said to be* ***(order)complete*** *iff for every non empty subsets $A, B \subseteq X$*

$$\forall a \in A \forall b \in B : a \leq b$$

$$\implies \exists c \in X \quad s.t.$$

$$\forall a \in A : a \leq c \land$$

$$\forall b \in B : c \leq b$$

Ex. 27. *The real numbers $\mathbb{R}$ satisfy axiomatic properties for addition, multiplication, distributivity and order. Furthermore $\mathbb{R}$ is an order complete field.*

Ex. 28. *$\mathbb{Q}$ is not order complete. Let $A = \{x \in \mathbb{Q} \mid x^2 \leq 2\}$ and $B = \{x \in \mathbb{Q} \mid x^2 \geq 2\}$. Then $\forall a \in A \forall b \in B : a \leq b$ but there is no $c \in \mathbb{Q}$ such that $c^2 = 2$.*

Thm. 29. *Every* ***non empty, bounded from above (below)*** *subset $X \subset \mathbb{R}$ has a* ***supremum (infimum)***.

Thm. 30. *The* ***Archimedian Principle*** *says that For $x \in \mathbb{R}$ and $y > 0$ there exists an element*

$$n \in \mathbb{N} \quad s.t. \quad n \cdot y > x$$

For every $\epsilon > 0$ there exists an element $n \in \mathbb{N}$ such that

$$\frac{1}{n} < \epsilon$$

Thm. 31. *The Young Inequality states that for all $\epsilon > 0$ and all $x, y \in \mathbb{R}$:*

$$2|xy| \leq \epsilon x^2 + \frac{1}{\epsilon} y^2$$

Vectors and Vector Spaces

Def. 32. *The* ***euclidian space*** *$\mathbb{R}^n$ is the set of all vectors $x = (x_1, x_2, \dots, x_n)$ where $x_i \in \mathbb{R}$.*

Def. 33. ***Addition*** *and* ***scalar multiplication*** *are defined as:*

$$x + y = (x_1 + y_1, x_2 + y_2, \dots, x_n + y_n)$$

$$\lambda x = (\lambda x_1, \lambda x_2, \dots, \lambda x_n)$$

Def. 34. *A* ***vector space*** *has to satisfy the following axioms: Commutativity:*

$$x + y = y + x$$

Associativity:

$$x + (y + z) = (x + y) + z$$

Identity element:

$$\exists e \in X \quad s.t. \; x + e = x \quad \forall x \in X$$

Inverse element:

$$\forall x \in X \exists x^{-1} \in X \quad s.t. \; x + x^{-1} = e$$

Distributivity:

$$x \cdot (y + z) = x \cdot y + x \cdot z$$

Def. 35. *For $x, y \in \mathbb{R}^n$, the* ***standard scalar product*** *is :=:*

$$x \cdot y = \langle x, y \rangle \coloneqq \sum_{i=1}^{n} x_i y_i$$

The ***euclidean norm*** *is:*

$$||x|| \coloneqq \sqrt{\sum_{i=1}^{n} x_i^2}$$

The ***euclidean distance*** *between x and y is:*

$$d(x, y) \coloneqq \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2}$$

Thm. 36. *The triangle- and the cauch-schwarz inequality hold for all $x, y, z \in \mathbb{R}^n$:*

$$||x - z|| \leq ||x - y|| + ||y - z||$$

$$|\langle x, y \rangle| \leq ||x|| ||y||$$

Complex Numbers

Def. 37. *The* ***complex numbers*** *$\mathbb{C}$ are the set of numbers*

$$\mathbb{C} = \{a + ib \mid a, b \in \mathbb{R}\}$$

where a is the ***real part*** *$Re(z)$, and b is the* ***imaginary part*** *$Im(z)$.*

Def. 38. *Arithmetic of complex numbers $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$: Addition:*

$$z_1 + z_2 = (x_1 + x_2) + i(y_1 + y_2)$$

Multiplication:

$$z_1 \cdot z_2 = (x_1 x_2 - y_1 y_2) + i(x_1 y_2 + y_1 x_2)$$

Def. 39. *The* ***complex conjugate*** *of $z = x + iy$ is $\bar{z} = x - iy$.*

Imp. 40. *Division is calculated by expanding with the conjugate of the denominator:*

$$\frac{z}{w} = \frac{z \cdot \bar{w}}{w \cdot \bar{w}} = \frac{z \cdot \bar{w}}{|w|^2}$$

Lem. 41. *Some usfull properties of the conjugate:*

$$z \cdot \bar{z} = |z|^2$$

$$z + \bar{z} = 2Re(z)$$

$$z - \bar{z} = 2iIm(z)$$

$$\overline{z + w} = \bar{z} + \bar{w}$$

$$\overline{z \cdot w} = \bar{z} \cdot \bar{w}$$

Def. 42. *Geometrically, complex numbers can be represented in the Gaussian plane. The* ***absolute value*** *of $z = a + ib$ is its* ***distance from the origin****:*

$$|z| = \sqrt{a^2 + b^2}$$

The distance between z_1 and z_2 is $d = |z_1 - z_2|$.

Thm. 43. *The Euler's formula states that for $t \in \mathbb{R}$:*

$$e^{it} = \cos(t) + i\sin(t)$$

This allows writing trigonometric functions using complex exponentials:

$$\cos(t) = \frac{e^{it} + e^{-it}}{2}$$

$$\sin(t) = \frac{e^{it} - e^{-it}}{2i}$$

Rem. 44. *The* ***Intuition*** *behind eulers formula is we can use Taylor series to show that:*

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

Using $x = it$ we get:

$$e^{it} = \sum_{n=0}^{\infty} \frac{(it)^n}{n!} = \cdots = \cos(t) + i\sin(t)$$

Def. 45. *For $z \in \mathbb{C}$, the polar-coordinates are defined as:*

$$z = r(\cos(t) + i\sin(t)) = re^{it}$$

where $r = |z|$ is the absolute value and $t = \arg(z)$ is the argument.

Rem. 46. *To convert from cartesian to polar coordinates:*

$$r = \sqrt{a^2 + b^2}$$

$$t = \arctan\left(\frac{b}{a}\right)$$

Note that the arctan function only returns values in $(-\frac{\pi}{2}, \frac{\pi}{2})$, so we have to adjust the angle based on the quadrant of the complex number.

Ex. 47. *Multiplication and division in polar-coordinates is much easier than in cartesian coordinates:*

$$z_1 \cdot z_2 = r_1 e^{it_1} \cdot r_2 e^{it_2} = r_1 r_2 e^{i(t_1 + t_2)}$$

$$\frac{z_1}{z_2} = \frac{r_1 e^{it_1}}{r_2 e^{it_2}} = \frac{r_1}{r_2} e^{i(t_1 - t_2)}$$

Ex. 48. *So is exponentiation or taking roots:*

$$z^n = (re^{it})^n = r^n e^{int}$$

$$\sqrt[n]{z} = \sqrt[n]{r} e^{i\frac{t + 2\pi k}{n}}, \quad k \in \{0, 1, \ldots, n-1\}$$

Def. 49. *The n-th roots of unity are the solutions to $z^n = 1$, which are given by:*

$$z_k = e^{i\frac{2\pi k}{n}}, \quad k \in \{0, 1, \ldots, n-1\}$$

Thm. 50. *The sum of the roots of unit is always zero.*

$$\sum_{k=0}^{n-1} e^{i\frac{2\pi k}{n}} = 0$$

Thm. 51. *The* ***fundamental theorem of algebra*** *states that every polynomial of degree $n \geq 1$ has exactly n roots in $\mathbb{C}$ (counting multiplicity). These roots appear in konplex-conjugate pairs.*

Sequences and Series

Sequences

Def. 52. *A* ***sequence*** *(Folge) $(a_n)_{n \in \mathbb{N}_0}$ is an infinite list of numbers, mathematically a mapping $f : \mathbb{N}_0 \to X$. It can be given explicitly or recursively as*

$$a_n = f(n) \quad \text{or} \quad a_n = g(a_{n-1}, a_{n-2}, \ldots)$$

Def. 53. *A sequence (a_n) is called* ***convergent*** *with limit $L \in \mathbb{R}$ if:*

$$\forall \epsilon > 0 \exists N > 0 \quad s.t.$$

$$\forall n > N : |a_n - L| < \epsilon$$

Written as $\lim_{n \to \infty} a_n = L$.If no such L exists, the sequence is ***divergent****.*

Rem. 54. *A special case of divergent sequences are sequences which diverge to $\pm\infty$. Their limits are called* ***improper limits****.*

Thm. 55. *The limit of a sequence is unique aslong as it exists.*

Thm. 56. *Any subsequence $(a_{n_k})_{k \in \mathbb{N}_0}$ of a convergent sequence converges to the same limit L.*

Def. 57. *An* ***accumulation point*** *$A \in \mathbb{R}$ of a sequence fulfills:*

$$\forall \epsilon > 0 \forall N \in \mathbb{N}_0 \exists n \geq N$$

$$|a_n - A| < \epsilon$$

Thm. 58. *A is an accumulation point $\iff$ there exists a convergent subsequence with limit A.*

Thm. 59. *Every convergent sequence has exactly one accumulation point (its limit).*

Def. 60. *A sequence is* ***upper (lower) bounded*** *if there exists M such that $|a_n| \leq (\geq) M \quad \forall n \in \mathbb{N}_0$.*

Thm. 61. *Every convergent sequence is bounded.*

Thm. 62. ***Bolzano-Weierstrass Theorem:*** *Every bounded sequence has an accumulation point.*

Thm. 63. *Limit arithmetic: For $\lim a_n = K$, $\lim b_n = L$ and $C \in \mathbb{R}$:*

$$\lim(a_n \pm b_n) = K \pm L$$

$$\lim(C \cdot a_n \cdot b_n) = C \cdot K \cdot L$$

$$\lim(a_n / b_n) = K/L$$

Rem. 64. *Important limits:*

$$\lim_{n \to \infty} \frac{\ln n}{n} = 0$$

$$\lim_{n \to \infty} n^{1/n} = 1$$

$$\lim_{n \to \infty} x^{1/n} = 1 \quad (x > 0)$$

$$\lim_{n \to \infty} \left(1 + \frac{x}{n}\right)^n = e^x$$

$$\lim_{n \to \infty} \frac{x^n}{n!} = 0$$

Thm. 65. ***Sandwich-Theorem:*** *If $a_n \leq b_n \leq c_n$ for $n \geq N_0$, and $\lim a_n = \lim c_n = L$, then $\lim b_n = L$.*

Def. 66. *A sequence is* ***monotonically increasing (decreasing)*** *if $a_n \leq (\geq) a_m$ for $m > n$.*

Thm. 67. ***Monotone Convergence Theorem:*** *Every bounded monotonic sequence converges to its supremum (infimum).*

Def. 68. *The limit of the supremum is called* ***limes superior*** *and denoted as:*

$$\limsup_{n \to \infty} a_n = \lim_{n \to \infty} \sup_{k \geq n} a_k$$

The limit of the infimum is called ***limes inferior*** *and denoted as:*

$$\liminf_{n \to \infty} a_n = \lim_{n \to \infty} \inf_{k \geq n} a_k$$

Def. 69. *A sequence is a* ***Cauchy sequence*** *iff*

$$\forall \epsilon > 0 \exists N \in \mathbb{N} \text{ such that}$$

$$\forall m, n \geq N : |a_n - a_m| < \epsilon$$

Thm. 70. *Every Cauchy sequence is bounded.*

Thm. 71. *A sequence of real numbers converges iff it is a cauchy sequence.*

Series

Def. 72. *A* ***series*** *is an infinite sum* $\sum_{i=0}^{\infty} a_i$ *of the terms of a sequence* (a_n).

Def. 73. *A series is said to converge iff the sequence of partialsums converges.*

$$\sum_{i=0}^{\infty} a_i = \lim_{n\to\infty} \sum_{i=0}^{n} a_i$$

Rem. 74. *Some examples of series are:* ***geometric series:***

$$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$$

for $|x| < 1$ ***harmonic series:***

$$\sum_{n=1}^{\infty} \frac{1}{n}$$

diverges for $s > 1$

Thm. 75. ***Necessary condition for convergence:*** *for* $\sum_{i=0}^{\infty} a_i$ *to converge we need* $\lim_{n\to\infty} a_n = 0$ *i.e. it is a null sequence.*

Thm. 76. *For* ***series with non-negative terms*** $a_n \geq 0$, *the sequence of partial sums is monotonically increasing and* ***converges iff it is bounded.***

Def. 77. *A series is* ***absolutely convergent*** *if* $\sum |a_n|$ *converges. A* ***conditionally convergent*** *series converges, but not absolutely.*

Thm. 78. ***Riemann's rearrangement theorem:*** *Let* $\sum a_n$ *be a conditionally convergent series. Then for any* $x \in \mathbb{R}$, *there exists a rearrangement of the series that converges to* x.

Thm. 79. ***Comparison test*** (Vergleichskriterium): *If* $c_n \leq b_n \leq a_n$ *for* $n \geq N$: *If* $\sum a_n$ *converges, then* $\sum b_n$ *converges* (***Majorantenkriterium***). *If* $\sum c_n$ *diverges, then* $\sum b_n$ *diverges* (***Minorantenkriterium***).

Thm. 80. ***Leibniz Criterion***: *For an alternating series* $\sum (-1)^n a_n$ *with a monotonically decreasing null sequence* (a_n), *the series converges.*

Thm. 81. ***Cauchy Criterion:*** *The series* $\sum a_n$ *converges iff*

$$\forall \epsilon > 0 \exists N \in \mathbb{N} \text{ such that}$$

$$\forall m, n \geq N : |\sum_{k=n}^{m} a_k| < \epsilon$$

Thm. 82. ***Root Criterion*** (Wurzelkriterium): *Let* $\rho = \limsup \sqrt[n]{|a_n|}$. *If* $\rho < 1$, *abs conv; if* $\rho > 1$, *div.*

Thm. 83. ***Ratio Criterion*** (Quotientenkriterium): *Let* $\rho = \lim \left|\frac{a_{n+1}}{a_n}\right|$. *If* $\rho < 1$, *abs conv; if* $\rho > 1$, *div.*

Def. 84. *A* ***power series*** (Potenzreihe) *centered at a has the form*

$$\sum_{k=0}^{\infty} c_k (x-a)^k$$

Thm. 85. *Every power series has a* ***convergence radius*** R, *with the* ***interval of convergence*** $(a-R, a+R)$. R *is computed by the formula* $R = \frac{1}{\limsup |c_n|^{1/n}}$.

Functions

Def. 86. *A* ***function*** $f : D \to C$ *is a map from a domian* D *to a codomain* C *s.t. for each* $x \in D$ *there is exactly one* $y \in C$ *st.* $f(x) = y$ *(well defined).*

Def. 87. *A function* f *is called* ***bouned*** *if there exists* $M \in \mathbb{R}$ *such that* $|f(x)| \leq (\geq) M$ *for all* $x \in D$.

Def. 88. *A function* f *is called* ***injective*** *iff*

$$\forall x_1, x_2 \in A : f(x_1) = f(x_2) \implies x_1 = x_2$$

surjective *iff*

$$\forall y \in B \exists x \in A : f(x) = y$$

and bijective iff f *is injective and surjective.*

Def. 89. *The* ***restriction*** *of* f *under* $D' \subset \mathbb{D}$:=

$$f \restriction_{D'} : D' \to C, x \mapsto f(x) \quad \forall x \in \mathbb{D}$$

Def. 90. *Let* $f : D \to C$ *and* $g : C \to B$ *be functions then*

$$g \circ f : D \to B \quad \text{with}$$

$$g \circ f(x) = g(f(x)) \quad \forall x \in D$$

is called the ***composition*** *of* f *and* g.

Def. 91. *The* ***identity function*** $id_D : D \to D := id_D(x) = x \quad \forall x \in D$.

Def. 92. *If a function* $f : D \to C$ *is bijective then there exists a unique function* $f^{-1} : C \to D$ *such that*

$$f(f^{-1}(y)) = id_C = y \quad \forall y \in C$$

and

$$f^{-1}(f(x)) = id_D = x \quad \forall x \in D$$

f^{-1} *is called the* ***inverse function*** *of* f.

Def. 93. *A function* $f : \mathbb{R} \to \mathbb{R}$ *is called* ***even*** *iff*

$$\forall x \in \mathbb{R} : f(-x) = f(x)$$

and ***odd*** *iff*

$$\forall x \in \mathbb{R} : f(-x) = -f(x)$$

Def. 94. *A function* $f : \mathbb{R} \to \mathbb{R}$ *is called* ***(strictly) monotonically increasing*** *iff*

$$\forall x_1, x_2 \in \mathbb{R} : x_1 < x_2 \implies f(x_1) < f(x_2)$$

and ***(strictly) monotonically decreasing*** *iff*

$$\forall x_1, x_2 \in \mathbb{R} : x_1 < x_2 \implies f(x_1) > f(x_2)$$

Thm. 95. *Every strictly monotonic function is injective.*

Limits in the context of functions

Def. 96. *Let* $f : \mathbb{D} \to \mathbb{R}$ *be a function satisfying* $\mathbb{D} \cap (x_0 - \delta, x_0 + \delta) \neq \emptyset$ *for all* $\delta > 0$ *and* $x_0 \in \mathbb{R}$. *We say that* $\lim_{x\to x_0} f(x) = L$ *iff*

$$\forall \varepsilon > 0 \exists \delta > 0 s.t.$$

$$x \in \mathbb{D} \cap (x_0 - \delta, x_0 + \delta) \implies |f(x) - L| < \varepsilon$$

The limit L *is unique.*

Def. 97. *A function* $f : \mathbb{D} \to \mathbb{R}$ *has in* x_0 *a* ***right-sided limit*** $\lim_{x \searrow x_0} f(x) = \lim_{x \to x_0^+} f(x) = L$ *iff*

$$\forall \varepsilon > 0 \exists \delta > 0 \quad s.t.$$

$$\forall x \in \mathbb{D}(x_0 < x < x_0 + \delta \implies |f(x) - L| < \varepsilon)$$

Def. 98. *A function* $f : \mathbb{D} \to \mathbb{R}$ *has a* ***limit for x approaching infinity*** $\lim_{x\to\infty} f(x) = L$ *iff*

$$\forall \varepsilon > 0 \exists M > 0 \quad s.t.$$

$$\forall x \in \mathbb{D}(x > M \implies |f(x) - L| < \varepsilon)$$

Def. 99. *A function* $f : \mathbb{D} \to \mathbb{R}$ *has in* x_0 *a* ***improper limit approaching infinity*** $\lim_{x\to x_0} f(x) = \infty$ *iff*

$$\forall N > 0 \exists \delta > 0 \quad s.t.$$

$$forall x \in \mathbb{D}(0 < |x - x_0| < \delta \implies f(x) > N)$$

Thm. 100. *Let* $\lim_{x\to x_0} f(x) = K (\neq \pm\infty)$ *and* $\lim_{x\to x_0} g(x) = L (\neq \pm\infty)$

$$\lim_{x\to x_0} (f(x) + g(x)) = K + L$$

$$\lim_{x\to x_0} (f(x) - g(x)) = K - L$$

$$\lim_{x\to x_0} (f(x)g(x)) = KL$$

$$\lim_{x\to x_0} (f(x)/g(x)) = K/L$$

$$\lim_{x\to x_0} (Af(x)) = AK$$

$$\lim_{x\to x_0} (f(x)/g(x)) = K/L if L \neq 0$$

Continuity of functions

Def. 101. *A function* $f : D \to \mathbb{R}$ *is called* ***continous*** *at* x_0 *if*

$$\forall \epsilon > 0 \exists \delta > 0 \text{ such that}$$

$$\forall x \in I(|\ x - x_0\ | < \delta \implies |\ f(x) - f(c)\ | > \epsilon)$$

The function is called continuous if it is continuous at every point in its domain.

Rem. 102. *The* ***intuition*** *behind this definition is that for any given distance* $\epsilon > 0$ *around* $f(x_0)$, *we can find a* $\delta > 0$ *such that for all* $x \in D$ *that are at least* δ *close to* x_0, *we have* $|f(x) - f(x_0)| < \epsilon$. *So for any bound on the output* (ϵ) *there exists a bound on the input* (δ) *such that all elements in the input bound* $(x_0 - \delta, x_0 + \delta)$ *are mapped to the output bound* $(f(x_0) - \epsilon, f(x_0) + \epsilon)$.

Def. 103. *Let* $f : D \to \mathbb{R}$ *be a function. Then* f *is called* ***left(right)-continuous*** *at* x_0 *if*

$$\lim_{x\to x_0^{-(+)}} f(x) = f(x_0)$$

Thm. 104. *Let* $f : D \subset \mathbb{D}(f) \subset \mathbb{R}$ *be a function and* $x_0 \in D$. f *is continuous at* x_0 *iff for every sequence* $(x_n)_{n\in\mathbb{N}} \subset D$ *with* $\lim_{n\to\infty} x_n = x_0$ *it holds that* $\lim_{n\to\infty} f(x_n) = f(x_0)$.

Rem. 105. *Continuity can also be viewed in terms of intervals, where a function* f *is continuous iff for every open interval* $I \subset \mathbb{R}$ *it holds that* $f^{-1}(I)$ *is open.*

Lem. 106. *Let* f, g *be continuous functions then*

$$f + g, \quad f - g, \quad f \cdot g, \quad c \cdot f, \quad \frac{f}{g}, \quad f \circ g$$

are continuous

Thm. 107. *Let I be an interval and let $f : I \to J$ be a continuous and bijective function. Then J is an interval too and the inverse function $f^{-1} : J \to I$ is continuous.*

Thm. 108. *Let $f : D \to \mathbb{R}$ be a continuous function. Let $c \in [f(a), f(b)]$ then there exists a $x_0 \in [a, b]$ such that $f(x_0) = c$.*

Lem. 109. *Let $[a, b]$ be a compact interval and let $(x_n)_{n\in\mathbb{N}} \subset [a, b]$ be a sequence. Then there exists a subsequence $(x_{n_k})_{k\in\mathbb{N}}$ such that*

$$\lim_{k\to\infty} x_{n_k} = x_0 \quad \text{for some } x_0 \in [a, b].$$

Def. 110. *Let $f : D \subset \mathbb{R} \to \mathbb{R}$ be a function If for $x_0 \in D$ it holds that $f(x_0) \geq f(x) \forall x \in D$, then f has a local maximum at x_0. If for $x_0 \in D$ it holds that $f(x_0) \leq f(x) \forall x \in D$, then f has a local minimum at x_0.*

Thm. 111. *Let $f : I \subset \mathbb{R} \to \mathbb{R}$ be a contինous function and I be a compact interval. Then f is bounded and attains its maximum and minimum value on I. Hence*

$$\exists x_{min}, x_{max} \in I \text{ such that}$$

$$f(x_{min}) \leq f(x) \leq f(x_{max}) \quad \forall x \in I$$

Differential calculus

Def. 112. *The **average rate of change** of a function f in the domain between a and $a + h$:=*

$$\frac{\Delta y}{\Delta x} = \frac{f(a + h) - f(a)}{h}$$

*and also called the **difference quotient**.*

Def. 113. *The **derivative** f' of a function f at the coordinate a :=*

$$\lim_{h\to 0} \frac{f(a + h) - f(a)}{h} = \lim_{b\to a} \frac{f(b) - f(a)}{b - a}$$

$$= \frac{df}{dx}|_{x=a} = f'(a)$$

> **Rem. 114.** *The derivative $f'(a)$ can geometrically interpreted as the incline/decline of the graph of f at the point $(a, f(a))$.*

Def. 115. *If for a function f at the coordinate a the derivative $f'(a)$ exists then f is called **differentiable** at a. If f is differentiable for every $a \in \mathbb{D}$ then f is called differentiable.*

Lem. 116. *If f is differentiable at x_0 then f is continuous at x_0.*

Def. 117. *If the limit*

$$\lim_{h>0\to 0} \frac{f(a + h) - f(a)}{h}$$

*exists then we call it the **right-sided derivative** (or the left-sided derivative analogue)*

Def. 118. *For a function f we can iteratively define higher derivatives for $n \in \mathbb{N}_0$*

$$f^{(1)} = f', f^{(2)} = f'', \dots f^{(n+1)} = (f^{(n)})'$$

*if $f^{(n)}$ exists we call it n**-times differentiable**.*

Def. 119. *if $f^{(n)}$ is continuous we call f n**-times continuous differentiable**. The set of all n-times continuous differentiable functions on D is denoted as $C^n(D)$*

Def. 120. *If for a function f and some $x_0 \in \mathbb{D}$ and $\delta > 0$ we have*

$$f(x_0) \geq (\leq) f(x) \quad \forall x \in \mathbb{D} \cap (x_0 - \delta, x_0 + \delta)$$

*then x_0 is called a **local maximum (minimum)**.*

Derivation rules

Thm. 121. *For functions $f(x)$ we have*

$$f(x) = ax^p \implies f'(x) = apx^{p-1}$$
$$f(x) = a^x \implies f'(x) = \ln(a) \cdot a^x$$
$$f(x) = \sin(x) \implies f'(x) = \cos(x)$$
$$f(x) = \cos(x) \implies f'(x) = -\sin(x)$$
$$f(x) = \tan(x) \implies f'(x) = \frac{1}{\cos^2(x)}$$
$$f(x) = \sinh(x) \implies f'(x) = \cosh(x)$$
$$f(x) = \cosh(x) \implies f'(x) = \sinh(x)$$
$$f(x) = \ln(x) \implies f'(x) = \frac{1}{x}$$
$$f(x) = \arcsin(x) \implies f'(x) = \frac{1}{\sqrt{1 - x^2}}$$
$$f(x) = \arccos(x) \implies f'(x) = -\frac{1}{\sqrt{1 - x^2}}$$
$$f(x) = \arctan(x) \implies f'(x) = \frac{1}{1 + x^2}$$

Thm. 122. ***Sum Rule:** For the n-th derivative of a sum:*

$$(f + g)^{(n)} = f^{(n)} + g^{(n)}$$

Thm. 123. ***General Leibniz Rule (Product):** The n-th derivative of a product is given by:*

$$(f \cdot g)^{(n)} = \sum_{k=0}^{n} \binom{n}{k} f^{(n-k)} g^{(k)}$$

Thm. 124. ***Chain Rule:** For the composition of functions:*

$$(f \circ g)'(x)$$
$$= f'(g(x)) \cdot g'(x)$$

Thm. 125. ***Quotient Rule:** For $g(x) \neq 0$:*

$$\left(\frac{f}{g}\right)'$$
$$= \frac{f'g - fg'}{g^2}$$

Thm. 126. ***Inverse Functions:** If f is differentiable and injective:*

$$(f^{-1})'(x)$$
$$= \frac{1}{f'(f^{-1}(x))}$$

Bernoulli-L'Hopital

Thm. 127. *Let $I \subset \mathbb{R}$ be an interval, $f : I \to \mathbb{R}$, and let x_0 be a local extremum of f. Then one of the following holds:*

1. *x_0 is an endpoint of I*
2. *f is not differentiable at x_0*
3. *f is differentiable at x_0 and $f'(x_0) = 0$*

Thm. 128. ***Rolle's Theorem**: Let $a < b$ with $f(a) = f(b)$, and let $f : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b). Then $\exists \xi \in (a, b)$ with*

$$f'(\xi) = 0$$

Thm. 129. ***Mean-Value-Theorem**: Let $a < b$, and let $f : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b). Then $\exists \xi \in (a, b)$ with*

$$f'(\xi) = \frac{f(b) - f(a)}{b - a}$$

Thm. 130. ***Cauchy's Mean-Value-Theorem**: Let $a < b$, and let $f, g : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b). Then $\exists \xi \in (a, b)$ with*

$$\frac{f(b) - f(a)}{g(b) - g(a)}$$
$$= \frac{f'(\xi)}{g'(\xi)}$$

Thm. 131. ***Bernoulli-L'Hopital's Rule:** Let $f, g : I \to \mathbb{R}$ be differentiable functions on (a, b) and let $x_0 \in (a, b)$. If $g(x) \neq 0$ and $g'(x) \neq 0$ for all $x \in (a, b)$. If*

$$\lim_{x\to x_0} |f(x)| = \lim_{x\to x_0} |g(x)| \in \{0, \infty\}$$

$$\text{and} \quad \exists L = \lim_{x\to x_0} \frac{f'(x)}{g'(x)}$$

then

$$\lim_{x\to x_0} \frac{f(x)}{g(x)} = \lim_{x\to x_0} \frac{f'(x)}{g'(x)} = L$$

Convex and Concave graphs

Def. 132. *A function $f : I \to \mathbb{R}$ is called **convex** iff $\forall a, b \in I \forall t \in (0, 1)$*

$$f((1 - t)a + bt) \leq (1 - t)f(a) + tf(b)$$

Def. 133. *A function $f : I \to \mathbb{R}$ is called **concave** iff $\forall a, b \in I \forall t \in (0, 1)$*

$$f((1 - t)a + bt) \geq (1 - t)f(a) + tf(b)$$

Thm. 134. *Let $(a, b) \subset \mathbb{R}$ and $f : (a, b) \to \mathbb{R}$ be differentiable. Then $\forall x$*

$$f'(x) \geq 0 \iff f \text{ is increasing}$$

Thm. 135. *Let $(a, b) \subset \mathbb{R}$ and $f : (a, b) \to \mathbb{R}$ be differentiable. Then*

$$f \text{ is convex} \iff f' \text{ is increasing}$$

Thm. 136. *Let $(a, b) \subset \mathbb{R}$ and $f : (a, b) \to \mathbb{R}$ be twice differentiable. Then*

$$f \text{ is convex} \iff f'' \geq 0$$

Taylor Series

Lem. 137. *We can locally approximate the graph of f at x_0 with the tangent line $t(x)$:*

$$f(x) \approx t(x)$$
$$= f(x_0) + f'(x_0)(x - x_0)$$

Lem. 138. *To even more accurately approximate the function f we can add further terms to the tangent line expansion:*

$$f(x) \approx t(x)$$
$$= f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2}(x - x_0)^2$$

Thm. 139. ***Taylor's Theorem:*** *Let $f : (a,b) \to \mathbb{R}$ be $n+1$ times differentiable. Then*

$$f(x) = P_n(x) + R(x)$$

with $P_n(x)$ the Taylor polynomial of degree n and $R(x)$ the remainder term:

$$P_n(x)$$

$$= \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x-x_0)^k$$

$$R(x)$$

$$= \frac{f^{(n+1)}(\xi)}{(n+1)!}(x-x_0)^{n+1} \quad \text{for some } \xi \in (x_0, x) \text{ or } (x$$

Def. 140. *The* ***Taylor series expansion*** *of an infinitely differentiable function $f : (a,b) \to \mathbb{R}$ at x_0 is given by:*

$$f(x)$$

$$= \sum_{k=0}^{\infty} \frac{f^{(k)}(x_0)}{k!}(x-x_0)^k$$

Thm. 141. ***Standard Taylor Series:*** *The following converge for all $x \in \mathbb{R}$:*

$$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

$$\sin x = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k+1}}{(2k+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$

$$\cos x = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k}}{(2k)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$$

Imp. 142. *Not all infinitely differentiable functions have a Taylor series expansion that converges to the function itself. For example, $\ln(1+x)$, $\frac{1}{1-x}$, and $(1+x)^p$ only converge for $-1 < x < 1$.*

Thm. 143. *Let $f(x) = \sum_{n=0}^{\infty} a_n x^n$ and $g(x) = \sum_{n=0}^{\infty} b_n x^n$ be two power series with radius of convergence R. Then:*

$$f(x)g(x)$$

$$= \sum_{n=0}^{\infty} \left(\sum_{k=0}^{n} a_k b_{n-k} \right) x^n$$

Thm. 144. *Let $f(x) = \sum_{n=0}^{\infty} a_n x^n$ be a power series with radius of convergence R. Then:*

$$f'(x)$$

$$= \sum_{n=1}^{\infty} n a_n x^{n-1}$$

Thm. 145. *Let $f(x) = \sum_{n=0}^{\infty} a_n x^n$ be a power series with radius of convergence R. Then:*

$$\int f(x)\,dx$$

$$= \sum_{n=0}^{\infty} \int a_n x^n \,dx$$

Integration

Indefinite Integrals and Techniques

Def. 146. *Functions $F : I \to \mathbb{R}$ such that $F'(x) = f(x)$ for all $x \in (a,b)$ are called* ***antiderivatives*** *of f.*

Def. 147. *The family of all anti-derivatives of a function $f : I \to \mathbb{R}$ for $I \subseteq \mathbb{R}$ an open interval is called the* ***indefinite integral*** *and :=*

$$\int f(x)\,dx$$

$$= \{F : I \to \mathbb{R} : F'(x) = f(x)\} = F(x) + C$$

Thm. 148. *Let $f, g : I \to \mathbb{R}$ be integrable functions and let $c \in \mathbb{R}$. Then:*

$$\int (f(x) \pm g(x))\,dx$$

$$= \int f(x)\,dx \pm \int g(x)\,dx$$

$$\int cf(x)\,dx = c \int f(x)\,dx$$

Thm. 149. ***Basic Antiderivatives:***

$$\int x^p\,dx = \frac{x^{p+1}}{p+1} + C \quad (p \neq -1)$$

$$\int \frac{1}{x}\,dx = \ln|x| + C$$

$$\int e^x\,dx = e^x + C$$

$$\int a^x\,dx = \frac{a^x}{\ln(a)} + C$$

$$\int \sin(x)\,dx = -\cos(x) + C$$

$$\int \cos(x)\,dx = \sin(x) + C$$

$$\int \frac{1}{1+x^2}\,dx = \arctan(x) + C$$

Thm. 150. ***Integration by Parts:*** *For differentiable functions $f(x)$ and $g(x)$ we have:*

$$\int f'(x)g(x)\,dx$$

$$= f(x)g(x) - \int f(x)g'(x)\,dx$$

Thm. 151. ***Substitution Rule:*** *For functions $f(x)$ and $g(x)$ we have:*

$$\int f(g(x))g'(x)\,dx$$

$$= \int f(u)\,du = F(g(x)) + C$$

where F is an antiderivative of f and $u = g(x)$.

The Definite Integral (Riemann Integral)

Def. 152. *Let U and L denote the upper and lower sums of a bounded function f. Then f is called* ***Riemann-integrable*** *iff $\inf U = \sup L$. The value of the definite integral $\int_a^b f(x)\,dx$ is then this common value.*

Thm. 153. ***Integrability Conditions:*** *Any continuous or monotone function on $[a,b]$ is Riemann-integrable.*

Thm. 154. ***Properties of Riemann Integral:*** *Let f, g be integrable and $c \in \mathbb{R}$:*

$$\int_a^b (f(x) \pm g(x))\,dx = \int_a^b f(x)\,dx \pm \int_a^b g(x)\,dx$$

$$\int_a^b cf(x)\,dx = c\int_a^b f(x)\,dx$$

$$f(x) \leq g(x) \implies \int_a^b f(x)\,dx \leq \int_a^b g(x)\,dx$$

$$\left| \int_a^b f(x)\,dx \right| \leq \int_a^b |f(x)|\,dx$$

$$\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx$$

$$\int_a^c f(x)\,dx = \int_a^b f(x)\,dx + \int_b^c f(x)\,dx$$

Thm. 155. ***Mean Value Theorem for Integrals:*** *Let $f : [a,b] \to \mathbb{R}$ be continuous. Then there exists a $c \in [a,b]$ such that:*

$$\int_a^b f(x)\,dx = (b-a)f(c)$$

Thm. 156. ***Fundamental Theorem of Integral Calculus:*** *Let $f : [a,b] \to \mathbb{R}$ be continuous. Then for all $C \in \mathbb{R}$, $F(x)$ is an antiderivative of f where:*

$$F(x) = \int_a^x f(y)\,dy + C$$

Sequences of Functions and Integrals

Def. 157. *Let $(f_n)_{n\in\mathbb{N}_0}$ be a sequence of functions.* ***Pointwise Convergence:*** *It converges pointwise to f if for every x, the sequence of real numbers $(f_n(x))$ converges to $f(x)$.*

Def. 158. ***Uniform Convergence:*** *It converges uniformly to f if for every $\epsilon > 0$, there exists an N such that for all $n \geq N$ and* ***for all*** *x simultaneously, $|f_n(x) - f(x)| < \epsilon$.*

Thm. 159. ***Swapping Limit and Integral:*** *If a sequence of integrable functions (f_n) converges* ***uniformly*** *to f on $[a,b]$, then the limit function f is also integrable, and:*

$$\int_a^b f(x)\,dx$$

$$= \lim_{n\to\infty} \int_a^b f_n(x)\,dx$$

Definite Integration Techniques

Lem. 160. *If $F : [a, b] \to \mathbb{R}$ is continuously differentiable, then:*

$$F(x) = F(a) + \int_a^x F'(t)\,dt$$

$$\int_a^b F'(t)\,dt = F(b) - F(a) = \Big[F(t)\Big]_a^b$$

Lem. 161. ***Substitution for Definite Integrals****: Let f be continuous and g continuously differentiable.*

$$\int_a^b f(g(x))g'(x)\,dx$$

$$= \int_{g(a)}^{g(b)} f(u)\,du$$

Version 2*: If $f'(x) \neq 0$ and f^{-1} is the inverse of f:*

$$\int_a^b g(f(x))\,dx$$

$$= \int_{f(a)}^{f(b)} g(y)(f^{-1})'(y)\,dy$$

Lem. 162. ***Integration by Parts for Definite Integrals****: Let f, g be continuously differentiable.*

$$\int_a^b f'(x)g(x)\,dx$$

$$= \Big[f(x)g(x)\Big]_a^b - \int_a^b f(x)g'(x)\,dx$$

Differential Equations

Modelling and Definitions

Rem. 163. ***Modelling Process****: To derive an ODE from a physical scenario:*

1. *Identify the quantity $u(t)$ and express its change over a small interval Δt: $u(t + \Delta t) = u(t) + (rate) \cdot \Delta t$*
2. *Rearrange and take the limit $\Delta t \to 0$: $\frac{u(t+\Delta t)-u(t)}{\Delta t} = rate \implies u'(t) = rate$*

Rem. 164. ***Typical ODE Models:***

1. *Bounded Growth (Logistic): With capacity L:*

$$u'(t) = ku(t)\left(1 - \frac{u(t)}{L}\right)$$

2. *Bimolecular Reaction ($A + B \to C$): With concentrations a, b:*

$$u'(t) = k(a - u(t))(b - u(t))$$

Def. 165. *An **ordinary differential equation (ODE)** contains derivatives of a function of one independent variable. It is:*

- ***of n-th order****: if the n-th derivative is the highest occurring.*
- ***linear****: if the unknown function $u(t)$ and its derivatives appear only as a linear combination.*
- ***homogeneous****: if it only contains terms with $u(t)$ and its derivatives (otherwise **inhomogeneous**).*

Thm. 166. ***Picard-Lindelöf (Existence and Uniqueness)****: If $f(x, y)$ is continuous and Lipschitz continuous with respect to y, then the IVP $y' = f(x, y)$ with $y(x_0) = y_0$ has a unique solution in some interval around x_0.*

First-Order ODEs

Thm. 167. *For an ODE of the form $y' = f(x)g(y)$, we can separate variables:*

$$\frac{dy}{dx} = f(x)g(y)$$

$$\implies \int \frac{1}{g(y)}\,dy = \int f(x)\,dx + C$$

Thm. 168. *If $y' = f(ax + by + c)$, substitute $u = ax + by + c$. Since $u' = a + by'$, this yields the separable ODE:*

$$u' = a + bf(u)$$

Thm. 169. *If $y' = g\left(\frac{y}{x}\right)$, substitute $u = \frac{y}{x}$. Since $y' = u'x + u$, this yields the separable ODE:*

$$u'x = g(u) - u$$

Linear ODEs of n-th Order

Thm. 170. ***Superposition Principle****: If y_1 and y_2 solve a linear homogeneous ODE, then for any $C_1, C_2 \in \mathbb{R}$:*

$$y(x) = C_1y_1(x) + C_2y_2(x)$$

Thm. 171. *An n-th order linear homogeneous ODE has n linearly independent solutions $y_1, \ldots, y_n$. Its general solution is:*

$$y(x) = \sum_{i=1}^{n} C_iy_i(x) \quad \text{with } C_i \in \mathbb{R}$$

Lem. 172. *For a homogeneous ODE with constant coefficients $a_ny^{(n)} + \cdots + a_0y = 0$, the Ansatz $y(x) = e^{\lambda x}$ yields the characteristic equation:*

$$a_n\lambda^n + a_{n-1}\lambda^{n-1} + \cdots + a_1\lambda + a_0 = 0$$

with characteristic polynomial $p(\lambda)$.

Thm. 173. ***General Solution from Characteristic Roots****: The roots of $p(\lambda)$ determine the fundamental solutions:*

- ***Distinct real root** $\lambda \in \mathbb{R}$*

$$y(x) = e^{\lambda x}$$

- ***Multiple real root** $\lambda \in \mathbb{R}$ **with multiplicity** m*

$$e^{\lambda x}, xe^{\lambda x}, \ldots, x^{m-1}e^{\lambda x}$$

- ***Complex pair** $\lambda = \alpha \pm i\beta \in \mathbb{C}$ **(multiplicity** m**)***

$$e^{\alpha x}\cos(\beta x), e^{\alpha x}\sin(\beta x), \ldots,$$

$$x^{m-1}e^{\alpha x}\cos(\beta x), x^{m-1}e^{\alpha x}\sin(\beta x)$$

Inhomogeneous Linear ODEs

Lem. 174. *The general solution of $a_ny^{(n)} + \cdots + a_0y = s(x)$ is $y(x) = y_h(x) + y_p(x)$, where y_h is the general homogeneous solution and y_p is a particular solution.*

Method of Undetermined Coefficients (Ansatz)

Thm. 175. ***Method of Undetermined Coefficients****: For right-hand side $s(x)$, choose the particular Ansatz $y_p(x)$ (without resonance):*

- ***Polynomial** $P_m(x)$:*

$$y_p(x) = C_mx^m + \cdots + C_1x + C_0$$

- ***Exponential** Ae^{kx}:*

$$y_p(x) = Ce^{kx}$$

- ***Trig** $A\sin(\omega x) + B\cos(\omega x)$:*

$$y_p(x) = C_1\sin(\omega x) + C_2\cos(\omega x)$$

Imp. 176. ***Resonance****: If the term e^{kx} or $e^{i\omega x}$ in $s(x)$ matches an m-fold root of the characteristic polynomial $p(\lambda)$, multiply the standard Ansatz $y_p(x)$ by x^m.*

Variation of Constants

Thm. 177. ***Variation of Constants (First-Order Linear ODE)****: For $y' + p(x)y = q(x)$, the general solution is:*

$$y(x) = e^{-P(x)}\left(\int q(x)e^{P(x)}\,dx + C\right)$$

where $P(x) = \int p(x)\,dx$ is an antiderivative of $p(x)$.

Initial Value Problems and Boundary-Value Problems

Def. 178. *An **Initial Value Problem (IVP)** specifies conditions at a single point x_0 (e.g. $y(x_0) = y_0$, $y'(x_0) = y_1$). A **Boundary Value Problem (BVP)** specifies conditions at multiple points (e.g. $y(a) = y_a$, $y(b) = y_b$).*